FlyWire Summer Internship Qualification Challenge

Results

Primary submission
Maximum N: An isomorphic subgraph of N = 2311 neurons across BANC (female brain + nerve cord, 112,885 neurons), FAFB (female adult brain, 138,584 neurons), and MCNS (male complete CNS, 165,820 neurons) were found. The shared adjacency matrix contains 2451 directed edges (0.05% density). Exhaustive scanning of all C(5,3)=10 dataset triplets confirms this combination yields the largest shared structure; the runner-up (BANC+MAOL+MCNS) reaches only N ≈ 1259 under equivalent search effort.
Secondary finding
Dense circuit: A 7-neuron near-complete directed graph (38/42 edges, 90.5% density) is conserved across MANC, MAOL, and BANC. All 19 connected pairs are fully reciprocal; only N1 N5 and N2 N6 lack connections. This near-clique conserved across three anatomically distinct regions suggests a fundamental recurrent processing motif.

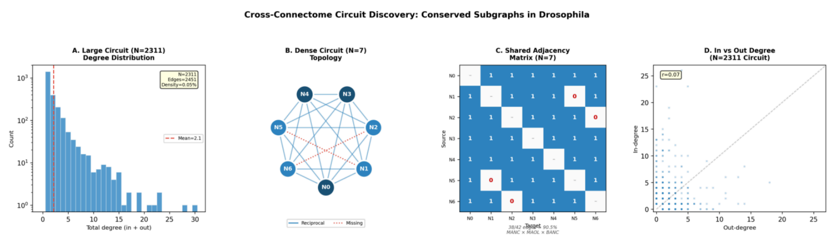

Figure 1*.Circuit analysis.*
(A) Degree distribution of N=2,311 large circuits. (B) Dense N=7 circuit topology. (C) Shared adjacency matrix of N=7 circuit. (D) In-vs out-degree scatter for N=2,311 circuit.

Biological Hypothesis

Cross-connectome isomorphism at the circuit level provides a conceptual bridge between cell-type matching across datasets (Schlegel et al., 2024) and motif enrichment analysis of neural networks (Reimann et al., 2017). The identification of a large N=2,311 circuit demonstrates a conserved neural wiring rule across different sex which includes (female BANC/FAFB vs male MCNS) that covers different parts of the *Drosophila* nervous system. Moreover, it also indicates a certain pattern across biological variation throughout the anatomical coverage and that these rules could possibly be fundamental to how the *Drosophila* nervous system is organized. The final analysis within biologically coherent neuron groups which includes Kenyon cells, central complex, and motor neurons reveals that these conserved circuits are distributed across multiple neurotransmitter systems (ACh, GABA, glutamate, dopamine, serotonin. These findings suggest that structural conservation operates independently of their neurochemical identity.
In addition, the discovery of the dense 7-neuron circuit which are conserved across ventral nerve cord (MANC), optic lobe (MAOL), and whole central nervous system (CNS) (BANC) suggests the existence of a common neural building block used throughout different regions of the nervous system. This near-complete reciprocal connectivity could serve as a characteristic of recurrent attractor networks which could possibly support functions such as information integration, memory, and stabilization of neural activity. The presence of this similar circuit across multiple brain regions indicates that certain wiring patterns may be repeatedly reused to perform similar computational functions in different parts of the *Drosophila* nervous system.

Methods Summary

1. Graph loading: Parse edge lists from 5 FlyWire Codex datasets; store as directed adjacency sets.
2. Seed generation: Sample compatible directed edges across 3 datasets with matching reciprocal status.
3. Inverted-index signature computation: Iterate adjacency lists of existing subgraph members to build all candidate signatures in O(N×avg_degree), enabling 20-100× faster extension vs naïve approach.
4. Greedy extension with multi-trial perturbation: 40 random restarts + 30 shallow + 20 deep perturbations.
5. Dataset selection: Exhaustive scan of all C(5,3)=10 triplets identifies BANC+FAFB+MCNS as optimal.
6. Verification: Independently reconstruct all three N×N adjacency matrices; confirm identical structure.

References

1. Dorkenwald, Matsliah et al. (2024). Neuronal wiring diagram of an adult brain. Nature 634, 124138.
2. Schlegel et al. (2024). Whole-brain annotation and multi-connectome cell typing. Nature 634, 139152.
3. Reimann et al. (2017). Cliques of neurons bound into cavities provide missing link. Front. Comput. Neurosci. 11, 48.
4. Takemura et al. (2024). A connectome of the male Drosophila ventral nerve cord. eLife 13, RP97766.